

HR Analytics - Employee Attrition Prediction

Abstract

This project analyzes employee attrition using HR data and machine learning techniques to identify factors influencing employee turnover and predict employees at risk of leaving.

Tools Used

Python, Google Colab, Pandas, NumPy, Seaborn, Matplotlib, Scikit-learn, Power BI.

Methodology

The dataset was cleaned, explored, and analyzed using EDA techniques. A Logistic Regression model was trained and evaluated using accuracy, classification report, and confusion matrix.

Key Findings

1. Overtime is associated with higher attrition.
2. Some job roles experience higher employee turnover.
3. Monthly income influences retention.
4. The model identifies employees at risk.
5. Risk segmentation supports retention strategies.

Conclusion

Data analytics and machine learning can help HR teams proactively identify attrition risks and improve workforce retention.